ISA/IEC 62443-3-3 / 4-2 • SL-3 TESTING CHEAT SHEET

Schneider Electric PowerLogic ION7650 Power & Energy Meter (ION7550/7650) · Component type: **Embedded Device (EDR) - power/energy (revenue/PQ) meter** · Target SL = SL 3

Where to work	**ION Setup** (Setup Assistant › **Security**: Standard vs **Advanced** security, up to **16 users** + access rights); the meter **front-panel display** (setup password + hardware/revenue lock); legacy embedded **web pages (HTTP)**; comms/protocol config in ION Setup. Firmware via ION Setup.
Ports / paths	Modbus/**TCP 502** (+ RTU); **DNP TCP 20000**; ION protocol **7700**; **SMTP 25** (email on events); legacy **HTTP 80**, **FTP 21**, **Telnet 23** (cleartext - disable); **SNMP 161**. **No HTTPS / SFTP** on this platform.
Notes	EDR - power/energy (revenue/PQ) meter, PowerLogic ION7550/ION7650. Confirm firmware and whether **Advanced security** is enabled (Standard = single meter password; Advanced = 16 users/roles). Legacy cleartext services must be disabled and the meter segmented. No secure-boot / HTTPS - those rows route to compensating controls. **Lost passwords are not recoverable** - custody matters.
How to read	Default read-only (ION Setup read-only, front-panel, read-only Modbus/DNP). Each row = native step-by-step (ION Setup / front panel) + command-level external-tool validation + pass. Write/exploit tools = isolated env + written authorization. Legacy meter - several rows are Partial/Compensating.

Project / config capture & evidence handling - capture the device backup + config export

Do this BEFORE working the checklist - screenshots and walking the config screens are not enough on their own. Capture the device's own backup / project export so nothing is missed:
1. Pull the native backup / project + full config export first (read-only, before anything): the ION Setup configuration + register framework export, the ION Setup security (users) settings, the front-panel hardware/revenue-lock state, the comms (ION/Modbus/DNP) + legacy-service (HTTP/FTP/Telnet) config, firmware rev, and the meter event log.
2. Export the security-relevant settings: user accounts / roles / password policy, comms / port / protocol enable-disable, and any certificate / key / CIP-Security settings - as files and screenshots.
3. Collect firmware & logs: firmware / boot revision, and the event / audit / diagnostic logs and fault / alarm buffers (exports) where the device provides them.
4. Capture the device identity: nameplate (catalog + serial), firmware / hardware rev, installed modules / option cards, and slot / network address - **photograph** the device, faceplate / HIM and panel labels.
5. Name + protect (keep everything): `<site>_<zone>_<assetTag>_<vendor>_<model>_<fwRev>_<YYYYMMDD-HHMM>_<initials>` with the native project / backup extension + the exports; SHA-256 hash the backup; store **only** on the encrypted evidence share; the backup may hold credentials / keys - **never** email it in cleartext and follow the project retention / classification policy. Do not save device passwords in tool profiles.
6. Then analyze offsite against the product-specific checklist, the product manuals, vendor research and experience; fill in the tool and auto-generate the report - and remember the manual fields the tool can't auto-write: the **executive-summary narrative**, the **overall assurance level + justification**, and the **punch-list**.

FR 1 — Identification & Authentication Control (IAC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 1.1	Human user identification & auth	In **ION Setup › Setup Assistant › Security** enable **Advanced security** so each of up to 16 users authenticates with a unique account + access rights (not a single shared meter password); set a strong front-panel setup password. Confirm no interface allows unauthenticated config. **Test:** attempt a setup change via ION Setup / front panel without credentials -> denied. **Ext. tools:** ION Setup (Security mode + user review); `nmap -p 502,80,21,23,20000,25 <ip>` (services in scope); check the legacy web login.	Advanced security on; per-user accounts; front-panel password; no unauthenticated config.	Advanced > Standard for SL-3.
SR/CR 1.2	Software process / device identification	Confirm a unique device identity: unique **IP address**, unique Modbus/ION unit ID, and meter serial; each meter distinct on the network. **Ext. tools:** `nmap --script modbus-discover -p 502 <ip>`; ION Setup device info; Wireshark `modbus || dnp3`.	Unique IP + Modbus/ION ID + serial; no reuse.	

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 1.3	Account management	In Advanced security review the up-to-16 user accounts + access rights - least privilege, named users, remove/rename any default, no shared logins. Note lost passwords require factory return, so manage custody carefully. **Ext. tools:** ION Setup (user/access-right export); credential-custody record.	Named, least-privilege accounts; defaults removed; custody controlled.	
SR/CR 1.4	Identifier management	Confirm unique user identifiers (Advanced security) and a unique device identity (IP/ION ID/serial); no reuse across meters. **Ext. tools:** ION Setup; `nmap --script modbus-discover -p 502`; inventory.	Unique user and device identifiers; no reuse.	
SR/CR 1.5	Authenticator management	Set strong per-user passwords (Advanced) and a strong front-panel password; change all defaults. Note: ION7650 password strength/policy is limited (legacy) - capture the enforced policy and compensate with segmentation + physical control. **Ext. tools:** ION Setup (password/policy review).	Strong non-default passwords set; policy captured; residual recorded if weak.	Partial (legacy policy).
SR 1.6	Wireless access management	**N/A.** Wired meter with no native wireless interface. **Ext. tools:** Confirm no wireless module.	N/A - justification recorded (wired device).	N/A
SR/CR 1.7	Strength of password-based authentication	Confirm the password-strength policy (length/complexity/lifetime) meets SL-3. **Steps:** for this product this is assessed with the same evidence as **SR/CR 1.5** above (shared steps); confirm this requirement individually and adjudicate it separately.	Strength of password-based authentication verified individually.	
SR/CR 1.8	Public key infrastructure (PKI) certificates	**N/A / limited.** The ION7650 has no device PKI/TLS (legacy HTTP/FTP/Telnet, no HTTPS/SFTP). Record N/A; provide confidentiality/integrity via segmentation and TLS at the SCADA/head-end layer. **Ext. tools:** Confirm no HTTPS (`nmap -p 443 <ip>` closed); Wireshark (web/FTP/Telnet = cleartext).	N/A/limited device PKI; compensating TLS/segmentation upstream.	N/A/Partial.
SR/CR 1.9	Strength of public key-based authentication	Confirm public-key authentication uses strong, current keys/algorithms. **Steps:** for this product this is assessed with the same evidence as **SR/CR 1.8** above (shared steps); confirm this requirement individually and adjudicate it separately.	Strength of public key-based authentication verified individually.	
SR/CR 1.10	Authenticator feedback	Confirm ION Setup, the front-panel, and the legacy web login mask password entry (no cleartext echo). **Ext. tools:** Visual verification; note Telnet/FTP transmit credentials in cleartext -> disable them.	Authenticators obscured at entry; cleartext-credential services disabled.	
SR/CR 1.11	Unsuccessful login attempts	ION7650 has **limited/no native lockout**; because lost passwords need a factory return, lockout is risky - compensate with segmentation, physical control, and monitoring rather than aggressive lockout. **Ext. tools:** Characterize behavior (authorized); IDS/SCADA on the meter segment for repeated auth failures.	Lockout where available; otherwise Partial + compensating controls.	Known limitation.
SR/CR 1.12	System use notification	No native banner; enforce the approved use notice on the ION Setup workstation (GPO) and any SCADA/head-end (compensating). **Ext. tools:** EWS GPO / SCADA banner review.	Use notice at EWS/SCADA (compensating at meter).	Compensating.
SR 1.13	Access via untrusted networks	Segment the meter; firewall **502/20000/80/21/23/25**; remote access via VPN/jump host; disable the cleartext services; no direct meter access from untrusted networks. Primary control for a legacy meter. **Test:** from an untrusted VLAN, reach 502 / the web -> must fail. **Ext. tools:** `nmap -Pn -p 502,80,21,23,20000,25,161 <ip>` from the untrusted side (expect filtered); firewall review.	Untrusted-network access denied; remote via VPN; cleartext services off.	

FR 2 — Use Control (UC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 2.1	Authorization enforcement	In Advanced security confirm per-user **access rights** (read vs configure vs reset) follow least privilege; the front-panel hardware lock and revenue seal restrict local changes. A read-only user cannot change setup or reset energy. **Test:** as a read-only user, attempt a setup change / energy reset -> denied. **Ext. tools:** ION Setup (access-right review); attempt a privileged action as a read-only user (isolated).	Least privilege enforced; privileged actions (config/reset) denied.	

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 2.2	Wireless use control	**N/A.** No wireless interface. **Ext. tools:** Confirm no wireless.	N/A - justification recorded.	N/A
SR 2.3	Use control for portable/mobile devices	Control the ION Setup laptop and any optical/serial config head; removable-media policy on the EWS; only authorized templates/firmware loaded. **Ext. tools:** EWS removable-media policy; physical inspection of the optical/serial port.	Portable/engineering media controlled.	
SR/EDR 2.4	Mobile code	No general-purpose mobile code; confirm only the ION framework configuration + firmware run and the legacy web content is vendor read-only. **Ext. tools:** ION Setup (framework review); browse the web read-only.	Only the trusted configuration/firmware runs; no untrusted mobile code.	EDR-specific.
SR/CR 2.5	Session lock	Confirm ION Setup and the front-panel return to a login/locked state after inactivity; the front panel re-prompts for the password. **Ext. tools:** ION Setup / front-panel idle test.	Sessions revert to login on inactivity.	Partial.
SR/CR 2.6	Remote session termination	Confirm sessions can be closed and idle sessions end; the meter bounds concurrent connections so simultaneous conflicting configuration is prevented. **Ext. tools:** ION Setup connection review; Wireshark (session teardown).	Sessions terminable; concurrency bounded.	
SR/CR 2.7	Concurrent session control	Confirm concurrent sessions are bounded/limited and monitored. **Steps:** for this product this is assessed with the same evidence as **SR/CR 2.6** above (shared steps); confirm this requirement individually and adjudicate it separately.	Concurrent session control verified individually.	
SR/CR 2.8	Auditable events	ION7650 has **no native syslog**. It keeps an on-meter event log / data recorders; capture setup changes and attribution in ION Setup and log meter/comms events at the SCADA/head-end + a network IDS; forward to the SIEM. Advanced security gives per-user attribution for setup changes. **Ext. tools:** ION Setup (event log / change review); SCADA/IDS/SIEM on the meter segment.	Setup changes attributable (Advanced security); events captured off-device.	No native syslog - Compensating.
SR/CR 2.9	Audit storage capacity	The on-meter event log / recorders are finite - offload to the SCADA/head-end/SIEM for retention and confirm alerting if collection fails. **Ext. tools:** SCADA/SIEM retention + source-silent alert; ION Setup (recorder depth).	Events offloaded/retained; audit-failure alerting defined.	
SR/CR 2.10	Response to audit processing failures	Confirm a defined response to audit-processing/forwarding failures (alerting; no silent loss). **Steps:** for this product this is assessed with the same evidence as **SR/CR 2.9** above (shared steps); confirm this requirement individually and adjudicate it separately.	Response to audit processing failures verified individually.	
SR/CR 2.11	Timestamps	Confirm the meter clock is synced (time sync via comms / optional GPS) so event-log/recorder timestamps are accurate and correlate. **Ext. tools:** ION Setup (time sync source); correlate with SCADA clock.	Reliable, synchronized timestamps.	
SR/CR 2.12	Non-repudiation	Confirm actions are attributable to a specific user (non-repudiation). **Steps:** for this product this is assessed with the same evidence as **SR/CR 2.8** above (shared steps); confirm this requirement individually and adjudicate it separately.	Non-repudiation verified individually.	
EDR 2.13	Physical diagnostic & test interfaces	Control physical interfaces: lock the enclosure; engage the **hardware/revenue lock** switch/link and anti-tamper seals; secure the optical/serial port; disable the legacy **web (HTTP), FTP, and Telnet** services in ION Setup. **Ext. tools:** Physical inspection (lock/seals); ION Setup (disable HTTP/FTP/Telnet); `nmap -p 21,23,80 <ip>` (closed).	Hardware lock/seals engaged; optical port + legacy services controlled/disabled.	EDR-specific.

FR 3 — System Integrity (SI)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 3.1	Communication integrity	ION/Modbus/DNP and the legacy web are **unauthenticated cleartext** and there is **no HTTPS/SFTP** - so integrity/confidentiality come from **containment**: disable HTTP/FTP/Telnet/SMTP if unused, keep Modbus/DNP/ION inside a segmented zone, and terminate TLS at the SCADA/head-end. No cleartext across a zone boundary. **Test:** confirm Modbus/DNP/web stay in-zone; FTP/Telnet disabled. **Ext. tools:** Wireshark `modbus || dnp3 || http || ftp || telnet` (cleartext, in-zone only); firewall review.	Control comms contained/segmented; cleartext services disabled; no crossing the boundary.	No HTTPS - Compensating.
SR/EDR 3.2	Protection from malicious code (component)	Confirm only Schneider firmware runs and there is no code path beyond the ION configuration. No signed-firmware guarantee is published for this legacy meter - verify firmware provenance from Schneider and control the update path tightly (2.3/3.10). **Ext. tools:** ION Setup (firmware version); Schneider provenance/hash; Schneider SEVD advisory.	Only vendor firmware runs; provenance verified; no code path.	EDR-specific/Partial.
SR/CR 3.3	Security functionality verification	After any change re-verify: Advanced security on, users/access rights, disabled legacy services (HTTP/FTP/Telnet/SMTP as unused), hardware lock/seals, and firmware, against the hardening baseline; compare the ION Setup configuration. **Ext. tools:** ION Setup (settings vs baseline); `nmap -p 21,23,80,161,502,20000,25 <ip>` (only required reachable).	Security-relevant settings match the baseline after change.	
SR/CR 3.4	Software & information integrity	Archive the ION Setup configuration/template, enable change control, and rely on Advanced security + hardware lock so unauthorized config/firmware change is detected. Compare online-vs-saved. **Ext. tools:** ION Setup (save/compare configuration); change-management record.	Config integrity controlled; unauthorized change detected.	Compensating.
SR/CR 3.5	Input validation	Confirm the meter tolerates malformed Modbus/DNP/ION frames without malfunction. **ISOLATED only:** fuzz the protocol interfaces on a bench meter and watch measurement/comms. **Ext. tools:** Modbus/DNP fuzzer (`boofuzz`) on a bench meter; Wireshark; watch the event log.	No malfunction on malformed input.	Do NOT fuzz an in-service revenue meter.
SR/CR 3.6	Deterministic output	Confirm defined behavior of any control relays / analog outputs and the data outputs on comm loss (last-known/defined state), and that measurement continues; this is primarily a measurement device. **Ext. tools:** ION Setup (output/relay + comm-loss behavior); simulate comm loss -> defined state.	Outputs/data assume a defined state on comm loss; measurement continues.	
SR/CR 3.7	Error handling	Confirm graceful diagnostics handling without info leak; review meter self-diagnostics / status. **Ext. tools:** ION Setup / front-panel (diagnostics/status review).	Graceful error handling; no unsafe state or info leak.	
SR/CR 3.8	Session integrity	**N/A / limited.** No secured (authenticated/encrypted) session (cleartext ION/Modbus/DNP, no TLS); integrity provided by containment. Record N/A + compensating segmentation. **Ext. tools:** Wireshark (confirm in-zone containment).	N/A/limited at device; compensating segmentation recorded.	N/A/Partial.
SR/CR 3.9	Protection of audit information	Since audit is off-device, protect the SCADA/SIEM store and the ION Setup config records; restrict who can clear on-meter logs (Advanced-security rights). **Ext. tools:** SCADA/SIEM access review; confirm clearing logs requires privilege.	Audit records protected from unauthorized read/modify/delete.	
EDR 3.10	Support for updates	Update firmware via ION Setup with firmware from Schneider; verify version/provenance; plan a maintenance window with rollback. (No on-meter signature verification - provenance discipline is the control.) **Ext. tools:** ION Setup (firmware update); Schneider provenance/version + SEVD advisory.	Controlled updates from a verified source, with rollback.	EDR-specific/Partial.
EDR 3.11	Physical tamper resistance/detection	Keep the meter in a locked panel; engage the **revenue/hardware lock** and anti-tamper seals; the event log records setup changes as tamper evidence; verify seals intact. **Ext. tools:** Physical inspection (lock/seals); ION Setup event-log review.	Lock/seals engaged; interfaces controlled; changes logged.	EDR-specific.
EDR 3.12	Provisioning product-supplier roots of trust	**Partial/N-A at device.** No signed-boot supplier root of trust on this legacy meter. Record the gap; compensate via firmware-provenance discipline (3.10), physical control, and segmentation. **Ext. tools:** Confirm no secure-boot/signed-firmware root; provenance review.	No documented supplier RoT; compensating controls recorded.	Partial/N-A.
EDR 3.13	Provisioning asset-owner roots of trust	**Partial/N-A at device.** No asset-owner certificate provisioning (no PKI/TLS). Record the gap; asset-owner trust enforced at the SCADA/network layer and by segmentation. **Ext. tools:** Confirm no device certificate capability; segmentation review.	No asset-owner RoT; compensating controls recorded.	Partial/N-A.

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
EDR 3.14	Integrity of the boot process	**Partial/N-A at device.** No published verified/secure boot. Record the gap; mitigate via firmware provenance (3.10), physical control, and monitoring. **Ext. tools:** Confirm no secure-boot capability; ION Setup firmware version.	No documented verified boot; compensating controls recorded.	Partial/N-A.

FR 4 — Data Confidentiality (DC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 4.1	Information confidentiality	Confirm information confidentiality — sensitive data is protected in transit and at rest. **Steps:** for this product this is assessed with the same evidence as **SR/CR 3.1** above (shared steps); confirm this requirement individually and adjudicate it separately.	Information confidentiality verified individually.	
SR/CR 4.2	Information persistence	On decommission/repurpose, reset the meter configuration to defaults (ION Setup) and clear user accounts so configuration/credentials are not recoverable; observe that lost passwords require factory reset anyway. **Ext. tools:** ION Setup (reset config / clear users).	Config and credentials purged on decommission.	
SR/CR 4.3	Use of cryptography	**N/A / limited.** No native crypto (cleartext protocols, no HTTPS). Disable unused services and rely on TLS at the SCADA/head-end and segmentation. **Ext. tools:** Confirm no device crypto; disable HTTP/FTP/Telnet/SMTP if unused.	N/A/limited; compensating TLS/segmentation upstream.	N/A/Partial.

FR 5 — Restricted Data Flow (RDF)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 5.1	Network segmentation	Primary control. Confirm the meter sits in a segmented metering/OT zone behind a firewall; only required protocols cross (Modbus 502 / DNP 20000 to authorized head-ends); legacy services blocked at the boundary; no flat convergence with IT; VPN for cross-zone. **Ext. tools:** `nmap -p 502,20000,80,21,23,25,161 <ip>` per zone; firewall rule review.	Meter in a segmented zone; only required flows cross the boundary.	
SR 5.2	Zone boundary protection	Confirm zone-boundary protection — only required flows cross the boundary (system SR / network device). **Steps:** for this product this is assessed with the same evidence as **SR/CR 5.1** above (shared steps); confirm this requirement individually and adjudicate it separately.	Zone boundary protection verified individually.	
SR 5.3	General-purpose person-to-person comms restrictions	Largely **N/A**, but note the meter can send **SMTP email (port 25)** on events; if unused disable it, or restrict SMTP so the meter originates no general-purpose messaging. **Ext. tools:** ION Setup (SMTP/email config) - confirm disabled/restricted.	N/A / SMTP email disabled or restricted.	N/A
SR 5.4	Application partitioning	**N/A.** Single-purpose meter; no separable applications to partition. **Ext. tools:** Confirm single-function device.	N/A - justification recorded.	N/A

FR 6 — Timely Response to Events (TRE)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 6.1	Audit log accessibility	Confirm the on-meter event log / recorders are readable/exportable (ION Setup) and forwarded to the SCADA/head-end/SIEM; a passive OT IDS (Modbus/DNP-aware) on the metering segment gives continuous monitoring (no native syslog). **Ext. tools:** ION Setup export; SCADA historian; passive OT IDS (Nozomi/Dragos) on a SPAN.	Events accessible/exportable and monitored in near-real-time.	No native syslog - Compensating.

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 6.2	Continuous monitoring	Confirm continuous monitoring — events forwarded to and monitored by a SIEM/OT-IDS in near-real-time. **Steps:** for this product this is assessed with the same evidence as **SR/CR 6.1** above (shared steps); confirm this requirement individually and adjudicate it separately.	Continuous monitoring verified individually.	

FR 7 — Resource Availability (RA)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 7.1	Denial-of-service protection	Confirm the meter tolerates Modbus/DNP scan floods without losing measurement; the cell firewall rate-limits. Legacy meters are sensitive to scan storms - monitor comms while testing. **Ext. tools:** ISOLATED bench: `hping3 --flood -p 502 <ip>` / Modbus storm; watch measurement + comms status.	Measurement continues under load; resources bounded.	
SR/CR 7.2	Resource management	Confirm bounded resources (connection limits) so a single service cannot starve the meter. **Ext. tools:** ION Setup / connection review; bench load test.	Resources bounded; no exhaustion-based loss of service.	
SR/CR 7.3	Control system backup	Confirm the ION Setup configuration/template is backed up (versioned, offline) and a restore is tested on a spare meter for a known-good rebuild. **Ext. tools:** ION Setup backup repository; restore test (spare meter).	Backups current and recovery validated.	
SR/CR 7.4	Control system recovery & reconstitution	Confirm recovery/reconstitution — a restore to a known-good state is tested. **Steps:** for this product this is assessed with the same evidence as **SR/CR 7.3** above (shared steps); confirm this requirement individually and adjudicate it separately.	Control system recovery & reconstitution verified individually.	
SR 7.5	Emergency Power	**N/A.** Emergency/backup power is provided at the facility/UPS level, not the device; confirm the device retains its security state across a power transfer.	N/A - justification recorded (not assessed as a control).	N/A
SR/CR 7.6	Network & security configuration settings	Harden: enable Advanced security, change defaults, engage the hardware lock, and **disable unused services** - HTTP web, FTP, Telnet, SMTP, SNMP, and any unused protocol - keeping only Modbus/DNP to authorized head-ends. **Ext. tools:** ION Setup (service/protocol config); `nmap -sTU -p 21,23,25,80,161,502,20000 <ip>` (only required reachable).	Least functionality; legacy/unused services disabled.	
SR/CR 7.7	Least functionality	Confirm least functionality — unused services/ports/protocols are disabled. **Steps:** for this product this is assessed with the same evidence as **SR/CR 7.6** above (shared steps); confirm this requirement individually and adjudicate it separately.	Least functionality verified individually.	
SR/CR 7.8	Component inventory	Confirm the meter (model ION7650, firmware, comms card) is inventoried so vulnerabilities track against Schneider SEVD / CISA advisories. **Ext. tools:** `nmap --script modbus-discover -p 502`; ION Setup inventory; Schneider SEVD feed.	Accurate meter/firmware inventory maintained.	

External Test Tools — how to use

Tool / method	Verifies	How to use — steps / example	Constraints
ION Setup	Native config & security review (all FRs)	Connect to the meter; Setup Assistant › Security (Standard/Advanced, users/access rights); communications (enable/disable HTTP/FTP/Telnet/SMTP/DNP/Modbus); event log; firmware. Save the configuration as the baseline.	Read/authorized; unlock only for change tests.
Front-panel display	Local access & lock review (1.1, 2.13)	Confirm the front-panel setup password, the hardware/revenue lock switch/link, and anti-tamper seals.	Physical; authorized.

Tool / method	Verifies	How to use — steps / example	Constraints
nmap	Service/port exposure, Modbus ID & segmentation (1.x, 3.1, 5.x, 7.6, 7.8)	`nmap -p 502,20000,80,21,23,25,161 <ip>`; `nmap --script modbus-discover -p 502 <ip>`; scan per zone; confirm FTP/Telnet/HTTP closed.	Read-only; authorized.
Wireshark / tcpdump	Comms integrity & identity (1.2, 3.1, 3.8)	Filter `modbus || dnp3 || http || ftp || telnet`; confirm cleartext stays in-zone and FTP/Telnet are disabled.	Passive; SPAN/tap only.
Modbus/DNP client (isolated)	Auth & authorization tests (1.2, 2.1)	From a non-authorized host attempt a Modbus/DNP write/reset -> confirm access rights block it; confirm read-only exposure otherwise.	ISOLATED for writes; authorized.
Modbus/DNP fuzzer / hping3	Input validation & DoS resilience (3.5, 7.1)	`boofuzz` Modbus/DNP on a bench meter; `hping3 --flood -p 502 <ip>` storm; watch measurement + comms.	ISOLATED bench; never in-service.
Passive OT IDS (Modbus/DNP-aware)	Monitoring (2.8, 6.x) - replaces native syslog	Nozomi/Dragos on a SPAN to capture Modbus/DNP connection/write and anomalies; forward to the SIEM.	Passive; coordinate with SOC.
Schneider SEVD / CISA advisories	Vuln tracking (3.2, 3.10, 7.8)	Track ION7550/ION7650 advisories and required firmware.	Reference.

Controlled - Project Use · Independent Assurance Services · ISA/IEC 62443-3-3 / 4-2 · Product-specific companion to the 62443 assessment tool; confirm exact menu paths / CLI against the deployed firmware and the project hardening baseline. ISOLATED tools run only in a representative/lab environment with written authorization.